Work Report (Week Aug 18)

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Experimental setup

The benchmark compares **five GNN training paradigms**:

- Central
- FedSage
- SplitFed
- PipeGCN-style
- SplitGNN-style

Each paradigm is trained using:

- **Single-Forward (SF):** layers are trained separately without backpropagation across layers.
- **Backpropagation (BP):** all layers are trained jointly from the final loss.

This produces **10 configurations**, evaluated over **5 runs** on Cora_ML using **four-layer GraphSAGE** and **five clients/workers**.

Main results

- **Central-SF achieved the highest average accuracy: 87.68%.**
- **PipeGCN-style-SF achieved 87.25%**, close to central training because it preserved all graph edges.
- **SplitGNN-style-SF achieved 85.74%** with very little communication.
- FedSage and SplitFed achieved approximately **68–72%** because their client partitioning retained only about 20% of the graph's edges.
- **SF outperformed its BP** counterpart in all five paradigms under the four-layer configuration.

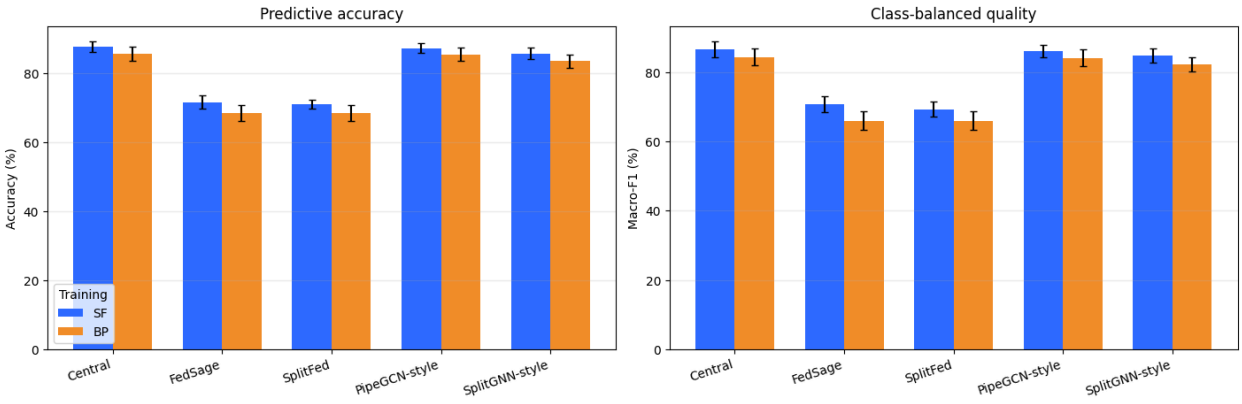


Fig: Test acc and F1 comparison across all experiments

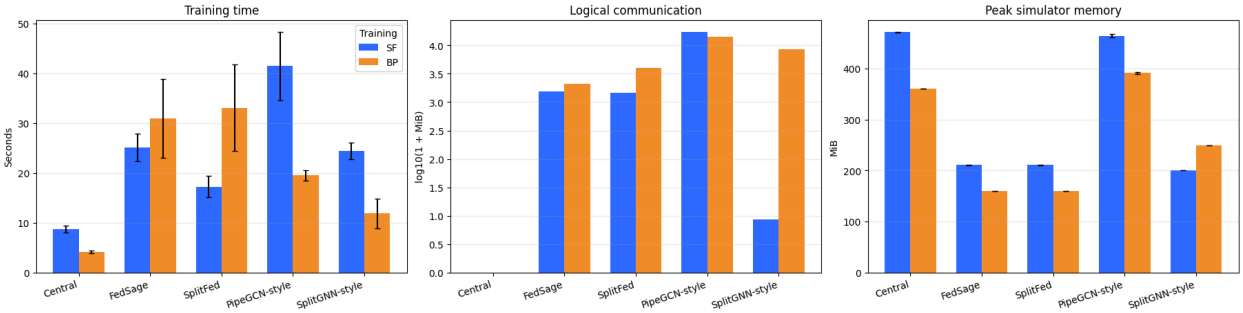


Fig: Training time, comm in data volume (logarithmic), Peak GPU usage

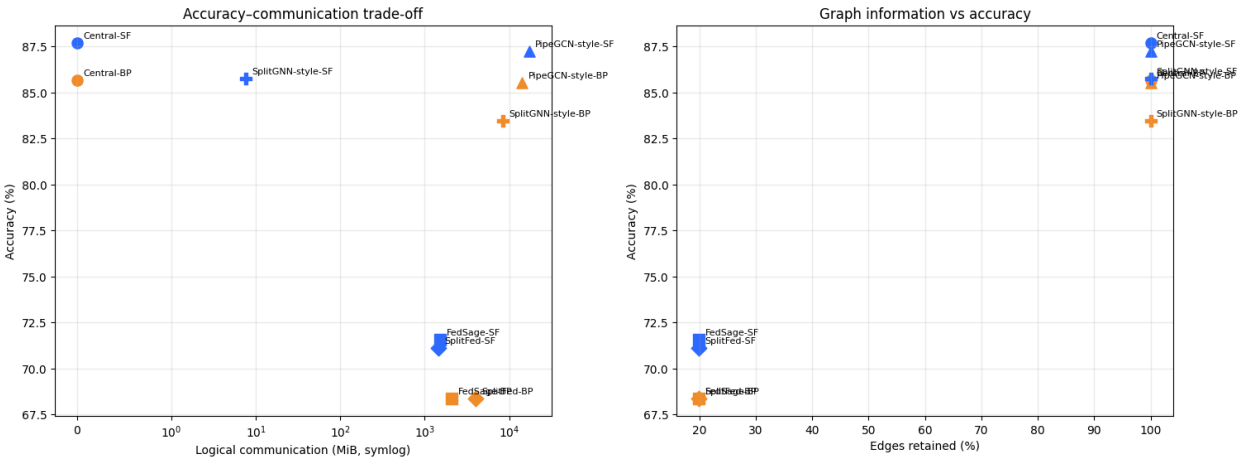


Fig: Trade-off charts

Main trade-offs

- ***Preserving cross-client edges*** generally produced better accuracy.
- FedSage and SplitFed ***reduced graph information*** and therefore lost accuracy.
- ***PipeGCN preserved accuracy but required the most communication.***
- ***SplitGNN-SF*** provided the ***best communication–accuracy balance***.
- **BP** was often **faster** because it trained the whole network once, while SF trained each layer separately.
- BP required repeated activation-gradient communication in split models, whereas SF could send detached embeddings only once.

Overall conclusion

The results show that BP-free Single-Forward training can remain competitive and sometimes outperform BP especially in deeper GNNs. However, distributed performance depends strongly on how much graph structure is preserved. The main challenge is therefore not only removing backpropagation, but also ***retaining important cross-client neighborhood information without excessive communication***.